

SWATI SHARMA

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EDUCATION

Indraprastha Institute of Information Technology (IIIT), Delhi, India

Bachelor of Technology in Computer Science and Artificial Intelligence

Coursework: Machine Learning, Deep Learning, Large Language Models, NLP, Computer Vision, Convex Optimization, Data Structures & Algorithms, Operating Systems, DBMS.

GPA: 8.36 / 10

Aug '21 – May '25

WORK EXPERIENCE

Google

Jun '25 – Present

Software Engineer II (Play Promotions)

Java, Kotlin, Python, GCL

- Led cross-team backend architecture for an in-app gift card promotion engine, designing cumulative gross-spend tracking primitives and policy governance rules to prevent promotion exploitation across global carts.
- Built Play Store promotional surfaces and quest banners in **Jetpack Compose** and extended backend schemas for **Google Play × Google Wallet** quests, boosting user enrollment by **20%** and generating **\$23M** in incremental revenue across **220M** Play Points members.
- Safely deprecated legacy post-purchase promotion infrastructure via shadow traffic (**~50 edge cases resolved**) and executed a feature-flagged migration of legacy campaign mappings to a scalable `offer_type` cache architecture, **cutting test runtime by 28%** and **flakiness by 42%**.
- Leveraged internal GenAI tooling (JetSki) to accelerate quest test development and offer onboarding at **10× velocity**, delivering org-wide tech talks on AI-assisted engineering.

NatWest Group

May '24 – Jul '24

Software Engineer Intern

AWS Inspector, S3, Docker, Nexus, Python

- Reduced **127 AWS Inspector vulnerabilities** by **56%** across the Kepler MLOps platform by hardening **8 Docker images**, optimizing base images, and enforcing least-privilege AWS S3/IAM policies.
- Modernized **6 ML product templates** (XGBoost, Decision Trees) by standardizing Docker and CI workflows, cutting pre-deployment security validation time by **30%**.

HMI Lab, IIIT Delhi [Web]

Jun '23 – Aug '23

Web Development Intern | [Dr. Jainendra Shukla](#)

HTML, CSS, JS

- Shipped a responsive lab website using modular CSS and deferred JS, **reducing load time by 30%** and achieving a **95% UX satisfaction score** across 50 surveyed members.

RESEARCH EXPERIENCE

Google DeepMind

Aug '25 – Present

Research Scientist (20% Policy) | [Dr. Vinay Namboodiri](#)

Python, JAX

- Developed **CRESS**, a CLIP-style multimodal dual-encoder (ResNet-18-D + SE, InfoNCE) over **1.06M** co-registered Sentinel-2 (10m) and aerial RGB (1m) pairs in **<30 mins (0.87 cosine similarity)**, formulating reference-free retrieval (**CRESS-R**) and latent Fréchet (**CRESS-C**) metrics to audit hallucinated structure in **10× super-resolution**.
- Enhanced **Mask2Former** for Satellite Image Time Series (SITS) field delineation with a temporal Exchanger backbone across **55K** images, resolving memory bottlenecks and **improving instance IoU by 0.31**.
- Built distributed **YOLOv11** training/inference pipelines in JAXline over **1.2M+** images for watershed instance separation, gaining **+0.09 instance IoU** and **cutting training time by 30%** via parallel prefetching.

Infosys Centre for AI, IIIT Delhi

Aug '23 – May '25

Undergraduate Researcher | [Dr. Anubha Gupta](#)

Python, PyTorch

- Built the **Task-Lens** cross-task utility profiling framework evaluating **50 Indian speech datasets (90K+ hours, 26 languages, 9 tasks)** to quantify metadata quality and cross-task transferability.

Human-Centered Design Lab, IIIT Delhi

Aug '23 – May '25

Undergraduate Researcher | [Dr. Jainendra Shukla](#)

Python, MATLAB

- Engineered an interpretable ML framework modeling temporal pupillometry dilation/constriction dynamics for clinical ADHD diagnosis, achieving **~90% classification accuracy**.

POSITIONS OF RESPONSIBILITY

- Reviewer**, WiML Workshop @ NeurIPS 2025
- Super Volunteer**, WiML Workshop @ NeurIPS 2025
- Mentor**, PIP (Interview Preparation), IIIT Delhi Jan '24 – May '25
- Member, **WiT (Women in Tech) & Girls' Residency**, IIITD 2023 – 2025

PUBLICATIONS & PRESENTATIONS

- [W.1] Sharma, S., Jayagopal, A., et al. (2026). **CRESS: Cross-Modal Reference-Free Evaluation for Remote Sensing Super-Resolution**. *NeurIPS Workshop on Tackling Climate Change with ML (TCCML) (Under Review)*.
- [C.1] Sharma, S., Sharma, D. V., & Gupta, A. (2026). **Task-Lens: Cross-Task Utility-Based Speech Dataset Profiling and Empirical Survey for Low-Resource Indian Languages**. *Language Resources and Evaluation Conference (LREC)*.
- [J.1] Sharma, S., Chakrabarty, M., Ray, S. B., & Shukla, J. (2025). **Machine Learning-Driven Analysis of Temporal Pupil Dynamics for Interpretable ADHD Diagnosis**. *Computers in Biology and Medicine*, 196(Pt C), 110878.
- [P.1] Sharma, S., et al. (2025). **Reference-Free Evaluation of RS Super-Resolution**. *Poster, Women in Machine Learning (WiML) Workshop @ NeurIPS*.

PROJECTS

Bridging OCR & Math Analysis [GitHub]

Jan '25 – May '25

Computer Vision | Python, Hugging Face

- Built a benchmark dataset and pipeline solving math problems from images via OCR, LaTeX reconstruction, and multi-step reasoning.
- Designed a structure-aware syntactic metric and evaluated OCR and vision-language models on real exam-style math images, exposing key failure modes.

Adaptive RL vs. Adversarial Attacks [GitHub]

Aug '24 – Dec '24

Large Language Models | Hugging Face, PyTorch, RLlib

- Designed an Adaptive RL framework using Proximal Policy Optimization (PPO) to detect and decrypt adversarial prompts, **reducing Attack Success Rate (ASR) by 30%**.
- Optimized defense efficiency with LoRA-based fine-tuning and 4-bit quantization, achieving a **96.8% Benign Success Rate**.

Review-Based Opinion Mining [GitHub]

Jan '24 – May '24

NLP | Python, PyTorch, Hugging Face, NLTK

- Detected fake reviews **15 pts** over chance on a balanced **21K-review** Amazon benchmark using a from-scratch encoder-only **Transformer** with self- and category cross-attention.
- Generated product descriptions from genuine reviews with **BART**, scoring review-description agreement via sentence-embedding cosine similarity.

CVD Risk Prediction Model [GitHub]

Aug '23 – Jan '24

Machine Learning | Python, Pandas, Scikit-learn

- Modeled individual patient Cardiovascular Disease risk using Survival Analysis, achieving a **mean C-index of 0.7585**.

TECHNICAL SKILLS

- Languages:** Python (NumPy, Pandas, Scikit-learn, Matplotlib), Java, Kotlin, C/C++ , SQL, JavaScript, HTML/CSS, MATLAB.
- ML/DL & GenAI:** PyTorch, JAX, TensorFlow, OpenCV, Hugging Face, vLLM, RLlib, LoRA/PEFT, OpenAI API.
- Cloud & Tools:** Kubernetes, Docker, Git, Bazel/Blaze, Protobuf, AWS (ECR, Inspector, S3, SageMaker), MySQL, MongoDB, Gradle.

AWARDS & ACHIEVEMENTS

- Grace Hopper Celebration (GHC) Scholar '24**; Microsoft Azure **#include <her> Achiever**.
- Top 2.5%** in Google Girl Hackathon 2023 [Learnings]; JEE Mains 2021: **99.67th Percentile (Top 0.34%)**.
- College Rank 1** in ProSort (IIIT Delhi Foobar); solved **600+** problems on [LeetCode](#), [Codeforces](#), [CodeChef](#), and [GFG](#).
- All India Rank 1 (AIR-1)**, National Interactive Math Olympiad (2018); **19 Gold Medals** in International Olympiads.